Md. Zahurul Islam

Ambagan, Jahangirnagar University, Savar, Dhaka -1342 |
(+880) 1634480532 | zahurulislamph@gmail.com | LinkedIn | **Portfolio** |

PROFESSIONAL SUMMARY

Geospatial Specialist with over 5 years of experience in GIS, remote sensing, spatial analysis, and WebGIS development, applied to environmental conservation, infrastructure planning, and archaeological research in Bangladesh. Skilled in managing end-to-end geospatial workflows, including GPS data collection, spatial database management, satellite image analysis, thematic mapping, and development of interactive web-based mapping applications using Leaflet and OpenLayers. Experienced in supporting government and international projects and translating complex spatial data into actionable insights for environmental management, heritage preservation, and development planning.

WORK EXPERIENCE

GIS Analyst

GeoSense Solutions Ltd. | Dhaka, Bangladesh | July 2024 – Present |

- Conduct spatial data analysis and geospatial modeling using ArcGIS Pro and QGIS to support environmental and development projects.
- Prepare thematic maps, spatial datasets, and analytical reports for project planning and decision-making.
- Manage geospatial databases and ensure spatial data quality and standardization.
- Support project teams with GIS-based visualization, spatial interpretation, and map production.

Junior Consultant (GIS & Spatial Monitoring)

Sustainable Forest & Livelihood (SUFAL) Project |
Bangladesh Forest Department (World Bank) | Dhaka, Patuakhali, Borguna, Bangladesh |
Sep 2023 – June 2024 | (10 months)

- Led GIS implementation for SMART patrolling systems to monitor forest resources and detect environmental threats.
- Designed spatial data models and managed patrol datasets for monitoring deforestation and wildlife risks.
- Conducted spatial and statistical analysis to identify environmental degradation patterns and support Site-Specific Planning (SSP).
- Trained forest department staff and local teams in GPS data collection, SMART monitoring tools, and GIS data management.
- Managed spatial databases and reporting systems integrated with national monitoring platforms.

Research Assistant & GIS Operator

Archaeological Survey of Dhaka Mass Rapid Transit (MRT Line-1 & 6)
Oitihya Onneswan - Archaeological Research Centre | Dhaka, Bangladesh |
Jan 2020 – Aug 2023 | (3.8 Years)

- Led GIS and field-based spatial analysis components of large-scale cultural and environmental surveys supporting national infrastructure development projects (Dhaka Mass Rapid Transit).

- Conducted raster and vector-based spatial analysis, land-use interpretation, and overlay mapping using ArcGIS and QGIS.
- Digitized and analyzed satellite imagery, historical maps, and cartographic records to support **heritage** site identification and planning decisions.
- Produced high-quality cartographic outputs and technical reports for government agencies, planners, and research stakeholders.

Research Assistant (GIS Mapping), Part-Time

Archaeobotanical Perspectives on Bangladesh (c. 500 BC – AD 1200) | **Archaeobotanical Project** |
University of Oxford | via Dr. Md. Mizanur Rahman | md.rahman@hertford.ox.ac.uk | Aug 2018 – Dec 2019

- Organized secondary spatial datasets, performed geo-referencing, related to archaeology, land use, and environmental history.
- Conducted GIS-based spatial analysis to identify land-use change patterns and site distributions over time.
- Integrated multi-source spatial data and produced interpretive thematic maps for publications and international presentations in collaboration with multidisciplinary research teams.

EDUCATION

Master of Science (M.Sc.) | 2019 | CGPA: 3.60/4.00
Bachelor of Science (B.Sc. Hons.) | 2018 | CGPA: 3.66/4.00
Geography & Environment
Jahangirnagar University, Dhaka, Bangladesh

RESEARCH & PUBLICATION

Islam, M. Z., Alim, A., & Moniruzzaman, S. (2024). Identifying Optimal Locations for Cyclone Shelters in underserved area in Kutubdia, Bangladesh : A GIS-Based Multi-Criteria Analysis. University of South Asia Journal (USAJ), 10(1), 1–12. ISSN: 2411-3077.
M.Sc. Thesis: Assessment of Landcover Changes in the Sundarban Mangrove Forest, Bangladesh (1990–2020) using RS and GIS *(April 2022).*

Technical SKILLS

- **Geospatial and Remote Sensing Tools:** ArcMap | ArcGIS Pro | QGIS | Google Earth Engine (GEE) | ERDAS Imagine |
- **WebGIS Development:** HTML5 | CSS3 | JavaScript | Leaflet (Learning) | OpenLayers (Learning) | GeoServer (Basic) | PostGIS (Learning) | PostgreSQL (Basic)
- **Spatial Analysis & GIS Techniques:** Spatial Analysis | Cartography | Land Use & Land Cover Analysis | Overlay Analysis | Geospatial Data Processing
- **Programming & Automation:** Python (Basic – geospatial data processing and automation) | GeoPandas (vector data analysis)
- **Data Analysis & Visualization:** Power BI | SPSS | ArcGIS StoryMaps
- **Other Tools:** MS Word | Excel | PowerPoint | Canva | VSDC Video Editor
- **Research:** Systematic Literature Review | Qualitative & Quantitative Analysis

TRAININGS & COURSES

Programming
Python Workshop for beginners (Ostad learning) | Python for Everyone (Esri).
GIS & Spatial Analysis
Esri Academy: 12 courses | Esri (2023–2025)
Key topics: Spatial Analysis | Climate Action | Land Cover Change Detection | Suitability Analysis | ArcGIS StoryMaps
Remote Sensing & Survey
Humanitarian Needs Assessment – KoboToolbox (2025) | Overview of GIS – ISRO (2024) | SuperMap GIS Training – SuperMap (2021) | Foundation Course in GIS, RS & Drone Studies – Jahangirnagar University (2019)
Computer
MS Office | SPSS, Photoshop – Jahangirnagar University (2019)
Professional Development
International Student Ambassador Training – Beihang University (2022)

LEADERSHIP & VOLUNTEER ACTIVITIES

Founder, Punduria Organization of Peace (POP) | Since 2015
A grassroots social organization focused on community development in rural Bangladesh.
Sports Leadership | 2015–2019
I Served as a Sports Secretary in the student council, captained department team and honored as a champion
Treasurer, Swadeshi Sangha | 2019–2021
Managed finances for Swadeshi Sangha, promoting local products and self-reliance through awareness campaigns and cultural events.
Vice President, Badhon (Blood Donation Organization), JU Zone **| 2016–2019**
International Student Ambassador, Beihang University | 2022.

REFERENCE

Md. Mehedi Hasan Khan
Assistant Project Director,
SUFAL Project, at Bangladesh Forest Department
Phone: +8801740974456 |
Email: mehedihasan69@gmail.com

Muhammad Mahbubul Alam
Assistant Director, Oitihya Onneswan
Phone: +8801612839150 |
Email: archaeologyresearch.bd@gmail.com

Muhammad Rezaul Rakib
Associate Professor, Department of Geography & Environment
Jahangirnagar University
Phone: +8801911636869 | **Email:** mrrakib@juniv.edu